

Chapter review and final check

Can you connect reproductive structures, cells and hormone feedback in a new case?

Learning goal

Explain normal reproductive pathways and use them to predict the effect of a changed structure or signal.

Before you begin

Return to the normal pathway before analysing a failure. A correct label is a starting point, not a complete explanation.

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Key terms: gamete · negative feedback · ovulation · corpus luteum

Vocabulary help

Key terms

Words for this lesson

Use these definitions when you need help with a term.

gamete

A reproductive cell containing one chromosome set.

negative feedback

A response that tends to oppose the initial change.

ovulation

Release of a secondary oocyte from a mature ovarian follicle.

corpus luteum

A temporary endocrine structure formed from follicular tissue after ovulation.

How to complete this section

- Use the comparisons to find the pathways you need to revisit.
- Follow the synthesis example. Then try the optional independent set before revealing feedback.
- Open the required final check and select Start. Correct six selections, save both written explanations, then select Submit and finish.
- Find first attempts and later work in All My Work.

Expected work: one completed final-check run. Textbook review and the independent set are optional.

Find the changed function before predicting the outcome

At this point, a list of structures is only the beginning of an explanation. Start a new case with the normal pathway. Then locate the changed step: producing a cell, supporting its development, moving it, adding fluid, releasing a hormone, or responding to that hormone. Two cases can have the same final outcome for different reasons.

Read the first comparison table one row at a time. The row tells you which function to compare; the two columns give examples from the male and female systems. Do not read the whole table as one continuous route. In both systems a gamete is a reproductive cell, not a hormone or a gland. The gonads, testes and ovaries, contain cells involved in gamete development and cells that release reproductive hormones.

Function	Male example	Female example
Gamete development	Seminiferous tubules, with Sertoli-cell support	Oocyte development within ovarian follicles
Transport	Epididymis → vas deferens → ejaculatory duct → urethra	Ovary releases oocyte; fimbriae help guide it into the oviduct
Hormone-producing tissue	Interstitial cells release testosterone	Follicle cells release estrogen; corpus luteum releases progesterone and estrogen
Central control	GnRH → anterior pituitary → FSH and LH	GnRH → anterior pituitary → FSH and LH
Useful caution	Sperm production is not accessory-gland secretion	Ovulation is not menstruation or implantation

Trace the sperm route before locating a blockage. Developing sperm form in seminiferous tubules of the testis with support from Sertoli cells. They then reach the epididymis, where they mature and are stored. From there the vas deferens carries them toward the ejaculatory duct and then the urethra. A block after the epididymis is a transport problem; it does not by itself prove that production in the testis has stopped.

Now separate sperm from semen. Semen contains sperm mixed with accessory-gland secretions. The seminal vesicles supply fluid containing fructose, a sugar that contributes energy for sperm activity. A reduced contribution of this fluid can therefore affect the conditions supporting sperm even when sperm are present and the transport ducts are open. It is not automatically evidence of failed chromosome reduction or absent testosterone.

In the female pathway, the ovary releases a secondary oocyte at ovulation. The fimbriae help guide it into an oviduct, where fertilization usually occurs. Early development then

moves toward the uterus; the blastocyst, a later early-embryo stage, can implant in its endometrium. The endometrium is the uterine lining. Ovulation, fertilization, implantation and menstruation are different events, at different sites or times.

Use those locations in an infection case. Scarring can narrow or block an oviduct and interrupt the route needed for sperm and an oocyte to meet or for early development to move toward the uterus. Ovarian hormone production can continue despite that damage. A lack of visible symptoms cannot establish that no infection is present. Explain the stated mechanism without making a personal diagnosis.

Connect each hormone to its source and responding cells

Both reproductive control pathways begin with the hypothalamus releasing gonadotropin-releasing hormone, GnRH. GnRH stimulates the anterior pituitary to release follicle-stimulating hormone, FSH, and luteinizing hormone, LH. These hormones travel in the blood to responsive cells. A hormone name alone is not the explanation: say which cells receive it and what they do.

In the testis, FSH acts on Sertoli cells, which support developing sperm within the seminiferous tubules. LH acts on interstitial cells, also called Leydig cells, stimulating testosterone production. Testosterone supports reproductive functions, including sperm development in the testis, and contributes inhibitory feedback to the hypothalamus and anterior pituitary. Sertoli-cell inhibin particularly restrains FSH release. Keep the two target-cell roles distinct: FSH does not make interstitial cells produce sperm.

Negative feedback opposes a change in a controlled system. If an outside testosterone signal increases inhibitory feedback, a functioning control system can reduce GnRH and pituitary stimulation. The testes may then produce less of their own testosterone, and support for sperm production can be disrupted. The external signal and the testes' own production are not the same measurement.

In the ovary, FSH supports follicular development, with LH also contributing to ovarian hormone production. The growing follicle releases estrogen. Sustained high estrogen before ovulation temporarily promotes positive feedback: pituitary LH release rises sharply. This LH surge helps trigger release of the secondary oocyte and formation of the corpus luteum from the follicular tissue that remains in the ovary.

The corpus luteum releases progesterone and estrogen. Progesterone helps maintain the endometrium during the luteal phase. This explains why a typical hormone graph has a broad progesterone rise after the brief LH surge: a new ovarian hormone-producing structure has become active. LH is not converted into progesterone in the blood. Without pregnancy support, the corpus luteum regresses, its hormone output falls and the functional lining is shed. Early-pregnancy hCG can maintain corpus-luteum support, as explained in lesson 09.

A target cell also needs a functioning receptor, a molecule that recognizes the hormone and enables its usual effect. A normal hormone concentration establishes availability, not a guaranteed response. If Leydig cells cannot respond to LH, more LH need not restore testosterone output. If endometrial cells cannot respond to progesterone, a normal

progesterone concentration need not produce the normal lining response. Distinguish a missing signal from a target that cannot use a present signal.

The second table applies this method to changed conditions. For each row, identify the normal step, locate the change and trace only the consequences supported by the case. Do not assume a duct blockage must stop all ovarian hormones, or that a later corpus-luteum failure proves earlier ovulation never happened. Likewise, a stated day in a model is evidence about that model, not a universal personal calendar.

Change	Normal-pathway reasoning
LH target cells fail to respond	Increasing LH cannot guarantee the usual target response.
Corpus luteum becomes inactive	Progesterone support falls; connect this with the endometrium.
Oviduct scarring obstructs transport	Cycles can remain normal while fertilisation becomes less likely.
No visible symptoms of infection	Symptoms alone cannot rule out an STI.
A model says day 14	Use the stated illustrative cycle; do not turn it into a universal calendar rule.

Worked example

Normal input but reduced pituitary response

Two otherwise comparable constructed anterior-pituitary tissue models receive the same pattern of brief GnRH pulses. Model A has functioning GnRH receptors and releases FSH and LH in response. Model B has non-functioning GnRH receptors and shows much less FSH/LH release. Does this show that a hypothalamus supplied too little GnRH? What would the reduced output mean for otherwise responsive gonadal target cells?

- Identify what is held equal: the GnRH signal supplied to the two tissues. Less GnRH was not supplied to B in this comparison. The experiment does not measure a real hypothalamus.
- Locate the changed step: the receptor response in the anterior-pituitary tissue. A signal can be available while the receiving cells cannot respond normally. The stated observations place the immediate difference at this target-response step.
- Trace the lower FSH/LH output onward. If those outputs reached otherwise responsive gonads, they would provide less normal stimulation of Sertoli cells and Leydig cells in the testis, or of follicular and other hormone-producing cells in the ovary.
- Separate signalling from transport: this model describes hormone reception and release. It gives no evidence of a physically blocked vas deferens or oviduct. A different mechanism would be needed to explain a duct obstruction.

- Check the conclusion: the tissue comparison supports a pituitary target-response explanation, not deficient GnRH production by a hypothalamus. It does not by itself predict every downstream hormone value or provide a personal diagnosis.

Try a new case

Try the whole set before viewing feedback. Record whether you used help. This optional formative check does not add to required progress.

A model testis has interstitial cells that cannot respond to LH. The hypothalamus and pituitary still respond normally to feedback. Which pattern is most consistent with this model?

- Reduced testosterone with increased stimulation from the control centres
- Reduced testosterone with permanently absent LH release
- Normal testosterone because FSH directly replaces LH at interstitial cells
- Increased testosterone with reduced stimulation from the control centres

A model shows an LH surge and release of an oocyte, followed by failure of the corpus luteum to remain active. Which change follows most directly?

- A direct blockage of the oviduct by falling progesterone
- Lower luteal progesterone and reduced support for the endometrium
- Persistently high luteal progesterone from the inactive corpus luteum
- Failure of the earlier LH surge to have occurred

A semen sample contains sperm and the transport ducts are open, but seminal-vesicle fluid contains unusually little fructose. Which function is affected most directly by this specific change?

- Testosterone synthesis in interstitial cells
- Chromosome reduction inside primary spermatocytes
- An energy contribution normally provided by an accessory-gland secretion
- The opening of the vas deferens during sperm transport

A tissue model receives a normal luteal progesterone concentration, but its endometrial cells have non-functioning progesterone receptors. Explain why hormone concentration alone cannot establish that the lining will respond normally. Distinguish hormone production from target-cell response.

[Write your response separately.]

Maximum 3200 characters. Writing is saved for review, not automatically graded.

- I used help before submitting.

Open required check · Chapter review and final check

Explain what you have learned

This check counts toward chapter progress. Correct the 6 selections to unlock 2 written explanations. Saving writing records your response; it does not grade its accuracy.

Required chapter check

Selection 1

Sperm production is normal, but a transport duct between the epididymis and ejaculatory duct is blocked. Which structure is affected?

- Vas deferens
- Seminiferous tubule
- Seminal vesicle
- Prostate gland

Selection 2

FSH is present, but Sertoli cells cannot respond to it. Which function is directly threatened?

- Fructose release by the sperm head
- Support of developing sperm
- Storage of urine in the ovary
- LH release from interstitial cells

Selection 3

Which structure remains in the ovary after ovulation and supports the endometrium through hormone release?

- Cervix
- Released secondary oocyte
- Corpus luteum
- Oviduct

Selection 4

An illustrative graph shows estrogen rising first, a short LH surge, and then a broad progesterone rise. What explains the last rise?

- Menstrual shedding produces progesterone
- Progesterone is converted directly from LH in blood
- The oocyte releases all progesterone before ovulation
- Hormone release by the newly formed corpus luteum

Selection 5

Outside testosterone increases feedback on a functioning hypothalamus and pituitary. Which response is most consistent with the model?

- Reduced upstream stimulation of the testis
- A physical blockage of every reproductive duct
- Conversion of testosterone into sperm
- A required increase in both FSH and LH

Selection 6

An STI causes scarring of an oviduct. Which explanation best connects this damage to reduced fertility?

- The cervix becomes a new sperm-production site
- The route for gamete encounter or transport can be disrupted
- The scar must directly remove all sex chromosomes
- The ovary must stop every hormone immediately

Answer the selections correctly to unlock the writing.

Written explanations

Written explanation 1

Compare the regulation of sperm production with the sequence leading to ovulation. Use GnRH, FSH, LH and the correct target cells or structures.

[Write your response separately.]

Maximum 3200 characters. Explain the biology in your own words.

Answer withheld for first attempt

Written explanation 2

In a hypothetical cycle, an LH surge is followed by ovulation, but the corpus luteum then fails to remain active. Explain the likely hormone and endometrial consequences. Distinguish this from an oviduct blockage.

[Write your response separately.]

Maximum 3200 characters. Explain the biology in your own words.

Answer withheld for first attempt

← STIs and reproductive function / Next: Overview →